

Nonparametrics

M514: Econometrics II

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References

- ▶ Nonparametrics, lecture slides, Bruce Hansen,
<https://users.ssc.wisc.edu/~bhansen/718/718.htm>

Kernel Density Estimation

Density Estimation

Let $X \in \mathbb{R}$ be a random variable with continuous distribution $F(x) = \mathbb{P}\{X \leq x\}$ and density $f(x) = \frac{d}{dx}F(x)$. The goal is to estimate $f(x)$ using iid samples $X_1, \dots, X_n$. A natural estimator for $F(x)$ is the empirical distribution function $\hat{F}(x)$ given by

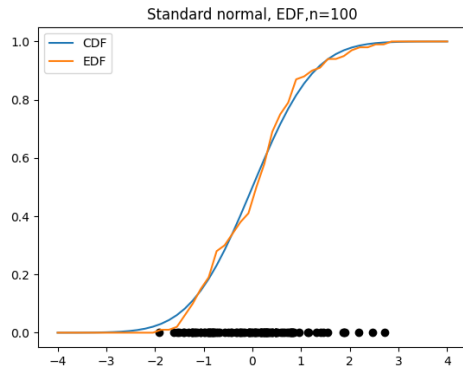
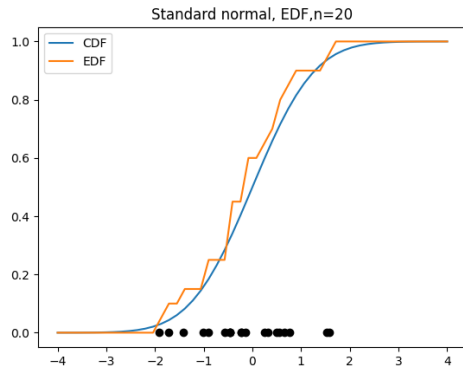
$$\hat{F}(x) = \frac{1}{n} \sum_{i=1}^n 1_{\{X_i \leq x\}}$$

This is a sample analog estimator because

$$\mathbb{E}1_{\{X_i \leq x\}} = \int_{\mathbb{R}} 1_{\{y \leq x\}} f(y) dy = \int_{-\infty}^x f(y) dy = \mathbb{P}\{X_i \leq x\}$$

where the first equality uses the law of unconscious statistician.

Density Estimation



The convergence of $\hat{F}(x)$ to $F(x)$ is actually uniform, i.e.,

$$\sup_{x \in \mathbb{R}} \left| \hat{F}(x) - F(x) \right| \rightarrow 0$$

as $n \rightarrow \infty$ by the Glivenko-Cantelli theorem.

Density Estimation

Using $f(x) = \frac{d}{dx}F(x)$, a natural estimator for $f(x)$ is given by the discrete derivative.

We define the discrete derivative as

$$\begin{aligned}\hat{f}(x) &= \frac{\hat{F}(x+h) - \hat{F}(x-h)}{2h} \\&= \frac{1}{2h} \left[\frac{1}{n} \sum_{i=1}^n 1_{\{X_i \leq x+h\}} - \frac{1}{n} \sum_{i=1}^n 1_{\{X_i \leq x-h\}} \right] \\&= \frac{1}{2nh} \sum_{i=1}^n [1_{\{X_i \leq x+h\}} - 1_{\{X_i \leq x-h\}}] \\&= \frac{1}{2nh} \sum_{i=1}^n 1_{\{x-h \leq X_i \leq x+h\}} \\&= \frac{1}{nh} \sum_{i=1}^n \frac{1}{2} \cdot 1_{\left\{ \frac{|X_i - x|}{h} \leq 1 \right\}}\end{aligned}$$

for $h > 0$ small.

Density Estimation

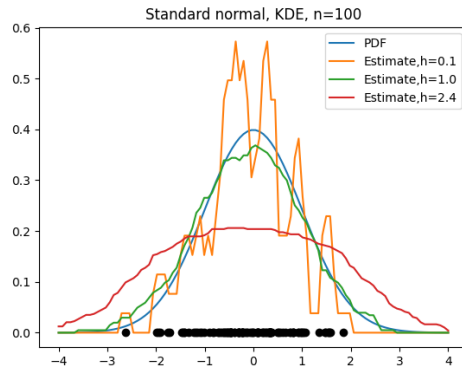
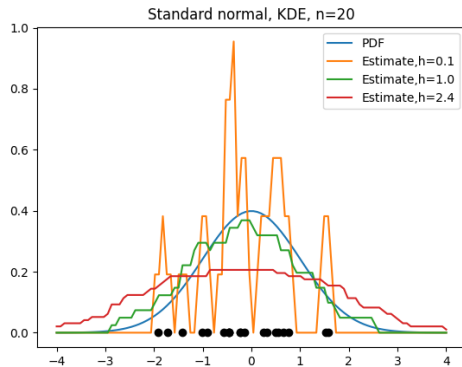
The discrete derivative estimator

$$\hat{f}(x) = \frac{1}{nh} \sum_{i=1}^n \frac{1}{2} \cdot 1_{\left\{\frac{|X_i - x|}{h} \leq 1\right\}}$$

counts how many samples fall within the interval $[x - h, x + h]$. If many samples are inside of the interval, $\hat{f}(x)$ will be large. Here h is a tuning parameter that we have to specify. How well the density estimator does will critically depend on the size of h .

1. Small h : Approximates the differentiation well (Low bias) but only small number of samples in the interval $[x - h, x + h]$ (High variance)
2. Large h : Many samples in the interval $[x - h, x + h]$ (Low variance) but doesn't approximate the partial derivative well (High bias).

Density Estimation



Kernel Density Estimation

Let

$$k(u) = \begin{cases} \frac{1}{2} & \text{if } |u| \leq 1 \\ 0 & \text{if } |u| > 1 \end{cases}$$

Then, the discrete derivative estimator

$$\hat{f}(x) = \frac{1}{nh} \sum_{i=1}^n \frac{1}{2} \cdot 1_{\left\{ \frac{|X_i - x|}{h} \leq 1 \right\}}$$

may be written as

$$\hat{f}(x) = \frac{1}{nh} \sum_{i=1}^n k\left(\frac{X_i - x}{h}\right)$$

Kernel Density Estimation

We may interpret the estimator

$$\hat{f}(x) = \frac{1}{nh} \sum_{i=1}^n k\left(\frac{X_i - x}{h}\right)$$

as computing weighted sum of 1's where the weights are given by $\frac{1}{h}k\left(\frac{X_i - x}{h}\right)$. The weighting function

$$k(u) = \begin{cases} \frac{1}{2} & \text{if } |u| \leq 1 \\ 0 & \text{if } |u| > 1 \end{cases}$$

is discontinuous at $|u| = 1$, causing abrupt changes in $\hat{f}(x)$. Can we do better?

Kernel Density Estimation

Let $k(u)$ be any function $k : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$\int_{-\infty}^{\infty} k(u) du = 1$$

We call such a function *kernel* function. We call a kernel function *symmetric* if it satisfies $k(-u) = k(u)$. We define the *order* of a kernel as its the first non-zero moment, i.e., the first integer j such that

$$\int_{-\infty}^{\infty} u^j k(u) du \neq 0$$

We mostly focus on the second order kernels, i.e, k such that

$$\int_{-\infty}^{\infty} uk(u) du = 0, \int_{-\infty}^{\infty} u^2 k(u) du > 0$$

Uniform Kernel

The function used in the discrete difference estimator

$$k(u) = \begin{cases} \frac{1}{2} & \text{if } |u| \leq 1 \\ 0 & \text{if } |u| > 1 \end{cases}$$

is a kernel because

$$\int_{-\infty}^{\infty} k(u) du = \int_{-1}^1 \frac{1}{2} du = 1$$

We may trivially see that it is symmetric. It is of second order because

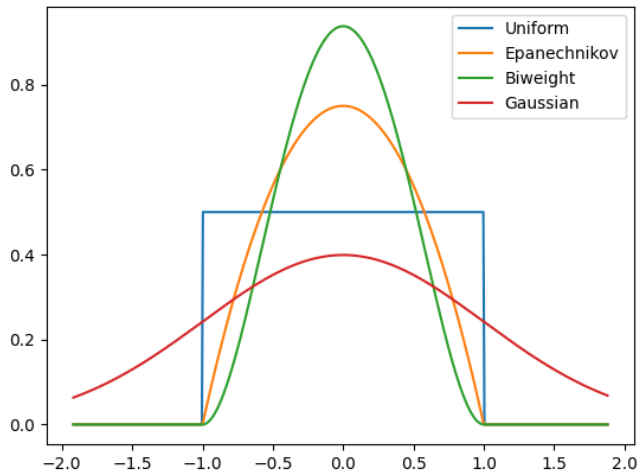
$$\begin{aligned} \int_{-\infty}^{\infty} uk(u) du &= \int_{-1}^1 \frac{1}{2} u du = 0 \\ \int_{-\infty}^{\infty} u^2 k(u) du &= \int_{-1}^1 \frac{1}{2} u^2 du = \frac{1}{2} \left(\frac{1}{3} + \frac{1}{3} \right) > 0 \end{aligned}$$

Kernels: Examples

Commonly used examples of second order kernel functions are:

Uniform	$k(u) = \frac{1}{2}1_{\{ u \leq 1\}}$
Epanechnikov	$k(u) = \frac{3}{4}(1 - u^2)1_{\{ u \leq 1\}}$
Biweight	$k(u) = \frac{15}{16}(1 - u^2)^2 1_{\{ u \leq 1\}}$
Triweight	$k(u) = \frac{35}{32}(1 - u^2)^3 1_{\{ u \leq 1\}}$
Gaussian	$k(u) = \frac{1}{\sqrt{2\pi}} e^{-\frac{u^2}{2}}$

Kernels: Examples



Kernel Density Estimation

The kernel density estimator given in the form

$$\hat{f}(x) = \frac{1}{nh} \sum_{i=1}^n k\left(\frac{X_i - x}{h}\right)$$

integrates up to one. To see this, first note that

$$\int_{-\infty}^{\infty} \frac{1}{h} k\left(\frac{X_i - x}{h}\right) dx = \int_{-\infty}^{\infty} k(u) du = 1$$

where we've used the change of variables $u = \frac{X_i - x}{h}$ with the Jacobian $\left|\frac{dx}{du}\right| = h$. Thus,

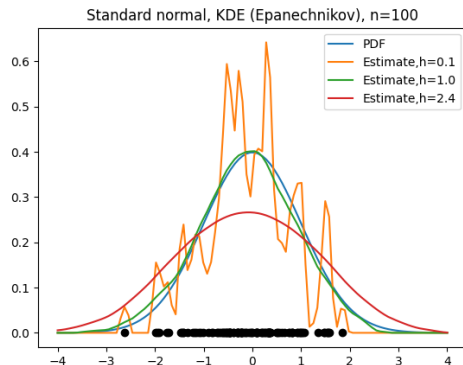
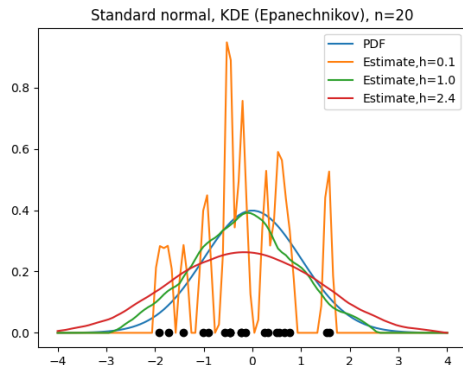
$$\begin{aligned} \int_{-\infty}^{\infty} \hat{f}(x) dx &= \int_{-\infty}^{\infty} \frac{1}{nh} \sum_{i=1}^n k\left(\frac{X_i - x}{h}\right) dx \\ &= \frac{1}{n} \sum_{i=1}^n 1 \\ &= 1 \end{aligned}$$

Kernel Density Estimation

The mean of the random variable X_i computed using the kernel density estimates $\hat{f}(x) = \frac{1}{nh} \sum_{i=1}^n k\left(\frac{X_i - x}{h}\right)$ is given by the sample mean. To see this,

$$\begin{aligned}\int_{-\infty}^{\infty} x \hat{f}(x) dx &= \frac{1}{n} \sum_{i=1}^n \int_{-\infty}^{\infty} x \frac{1}{h} k\left(\frac{X_i - x}{h}\right) dx \\&= \frac{1}{n} \sum_{i=1}^n \int_{-\infty}^{\infty} (X_i + uh) k(u) du \\&= \frac{1}{n} \sum_{i=1}^n X_i \int_{-\infty}^{\infty} k(u) du + \frac{1}{n} \sum_{i=1}^n h \int_{-\infty}^{\infty} u k(u) du \\&= \frac{1}{n} \sum_{i=1}^n X_i\end{aligned}$$

Density Estimation



Using Epanechnikov kernel, the estimator $\hat{f}(x)$ is more smooth. Is this always good? It is well known that the choice of the bandwidth parameter h is more important than the choice of the kernel k .

How to choose h ?

- ▶ **Theory:** It is optimal, in the sense of minimizing the MSE, to choose h in the form $Cn^{-1/(2j+1)}$ for some constant C , where j is the order of the kernel function. But how to choose/estimate C ?
- ▶ **Practice:** We may use the Silverman's rule-of-thumb method to choose h (This gives heuristics to choose C). It is always good to plot $\hat{f}(x)$ to see if it looks reasonable.

What if we have $X_i \in \mathbb{R}^k$?

Suppose we want to estimate a density function of a multivariate random variable $X_i = (X_{i1}, \dots, X_{ik})'$. All the previous arguments still apply with the univariate kernel function $k : \mathbb{R} \rightarrow \mathbb{R}$ replaced with a multivariate kernel function $K : \mathbb{R}^k \rightarrow \mathbb{R}$ that satisfies

$$\int_{\mathbb{R}^k} K(u) du = \int_{\mathbb{R}^k} K(u_1, \dots, u_k) du_1 \cdots du_k = 1$$

and the kernel density estimator given as

$$\hat{f}(x) = \frac{1}{nh^k} \sum_{i=1}^n K\left(\frac{X_i - x}{h}\right) = \frac{1}{nh^k} \sum_{i=1}^n K\left(\frac{X_{i1} - x_1}{h}, \dots, \frac{X_{ik} - x_k}{h}\right)$$

Typically, the multivariate kernel function takes the product form:

$$K(u) = k(u_1)k(u_2) \cdots k(u_k)$$

Nonparametric Regression

Nonparametric Regression

Let the data be given as $(Y_i, X_i), i = 1, \dots, n$ where $Y_i \in \mathbb{R}$ and $X_i \in \mathbb{R}$. The goal is to estimate the conditional expectation function $m : \mathbb{R} \rightarrow \mathbb{R}$ defined as

$$m(x) := \mathbb{E}[Y_i | X_i = x]$$

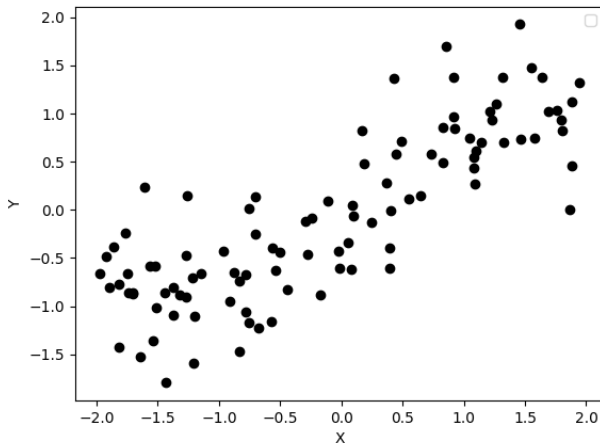
We may write

$$\begin{aligned} Y_i &= m(X_i) + e_i \\ \mathbb{E}[e_i | X_i] &= 0 \end{aligned}$$

How to estimate $m(x)$?

Local Constant (Nadaraya-Watson) Estimator

Fixing the value at which we want to evaluate m at x_0 , what would be a natural way of estimating $m(x_0)$?



Local Constant (Nadaraya-Watson) Estimator

Fixing $h > 0$, we may estimate $\hat{m}(x_0)$ as the average of Y_i around the samples near x_0 , i.e.,

$$\hat{m}(x_0) = \frac{\sum_{i=1}^n 1_{\{|X_i - x_0| \leq h\}} Y_i}{\sum_{i=1}^n 1_{\{|X_i - x_0| \leq h\}}}$$

Local Constant (Nadaraya-Watson) Estimator: Intuition

If the regression function $m(x)$ has bounded derivative, it should be reasonably constant over the interval $[x_0 - h, x_0 + h]$ (if h is small enough), so we can take the average the Y_i values for these observations.

$$\begin{aligned}\hat{m}(x_0) &= \frac{\sum_{i=1}^n 1_{\{|X_i - x_0| \leq h\}} Y_i}{\sum_{i=1}^n 1_{\{|X_i - x_0| \leq h\}}} \\&= \frac{\sum_{i=1}^n 1_{\{|X_i - x_0| \leq h\}} m(X_i)}{\sum_{i=1}^n 1_{\{|X_i - x_0| \leq h\}}} + \frac{\sum_{i=1}^n 1_{\{|X_i - x_0| \leq h\}} e_i}{\sum_{i=1}^n 1_{\{|X_i - x_0| \leq h\}}} \\&= m(x_0) + \frac{\sum_{i=1}^n 1_{\{|X_i - x_0| \leq h\}} (m(X_i) - m(x_0))}{\sum_{i=1}^n 1_{\{|X_i - x_0| \leq h\}}} + \frac{\sum_{i=1}^n 1_{\{|X_i - x_0| \leq h\}} e_i}{\sum_{i=1}^n 1_{\{|X_i - x_0| \leq h\}}}\end{aligned}$$

Local Constant (Nadaraya-Watson) Estimator: Intuition

Thus, using triangle inequality, we have

$$|\hat{m}(x_0) - m(x_0)| \leq \left| \frac{\sum_{i=1}^n 1_{\{|X_i - x_0| \leq h\}} (m(X_i) - m(x_0))}{\sum_{i=1}^n 1_{\{|X_i - x_0| \leq h\}}} \right| + \left| \frac{\sum_{i=1}^n 1_{\{|X_i - x_0| \leq h\}} e_i}{\sum_{i=1}^n 1_{\{|X_i - x_0| \leq h\}}} \right|$$

The mean-value theorem tells us that there exists some $\tilde{x}$ between X_i and x_0 such that

$$m(X_i) - m(x_0) = m'(\tilde{x})(X_i - x_0)$$

Note that for $X_i \in [x_0 - h, x_0 + h]$, we have

$$\begin{aligned} |m(X_i) - m(x_0)| &= |m'(\tilde{x})(X_i - x_0)| \\ &\leq |m'(\tilde{x})| h \\ &\leq \left(\sup_x m'(x) \right) h \end{aligned}$$

so that we may upper bound the first term using $(\sup_x m'(x)) h$.

Local Constant (Nadaraya-Watson) Estimator: Intuition

The second term

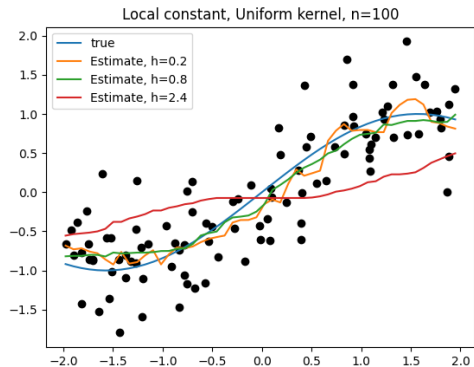
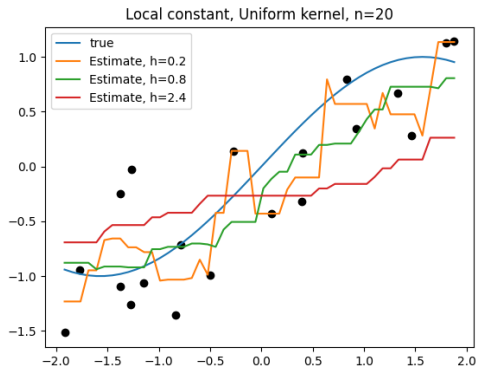
$$\left| \frac{\sum_{i=1}^n 1_{\{|X_i - x_0| \leq h\}} e_i}{\sum_{i=1}^n 1_{\{|X_i - x_0| \leq h\}}} \right|$$

is sample average of e_i for $X_i \in [x_0 - h, x_0 + h]$. This is an estimator for $\mathbb{E}[e_i | X \in [x_0 - h, x_0 + h]]$ and converges to $\mathbb{E}[e_i | X \in [x_0 - h, x_0 + h]] = 0$ if the number of X_i 's in $[x_0 - h, x_0 + h]$ is large.

Same intuition as before:

1. Large h : $|m(X_i) - m(x_0)|$ is large $\implies$ Large bias
2. Small h : Number of samples in $[x_0 - h, x_0 + h]$ is small $\implies$ Large variance

Local Constant (Nadaraya-Watson) Estimator



Local Constant (Nadaraya-Watson) Estimator

As with discrete derivative density estimator, we may write our estimator in terms of uniform kernel.

$$\hat{m}(x_0) = \frac{\sum_{i=1}^n 1_{\{|X_i - x_0| \leq h\}} Y_i}{\sum_{i=1}^n 1_{\{|X_i - x_0| \leq h\}}} = \frac{\sum_{i=1}^n k\left(\frac{X_i - x_0}{h}\right) Y_i}{\sum_{i=1}^n k\left(\frac{X_i - x_0}{h}\right)}$$

where

$$k(u) = \begin{cases} \frac{1}{2} & \text{if } |u| \leq 1 \\ 0 & \text{if } |u| > 1 \end{cases}$$

Local Constant (Nadaraya-Watson) Estimator

This estimator is a weighted average of Y_i , written in the form

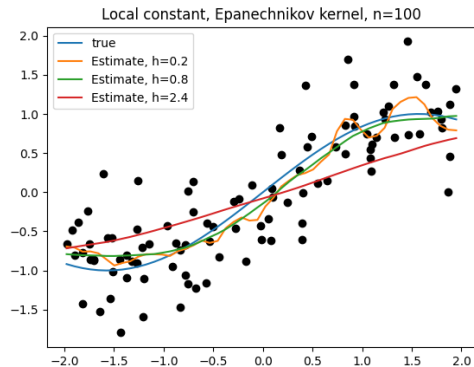
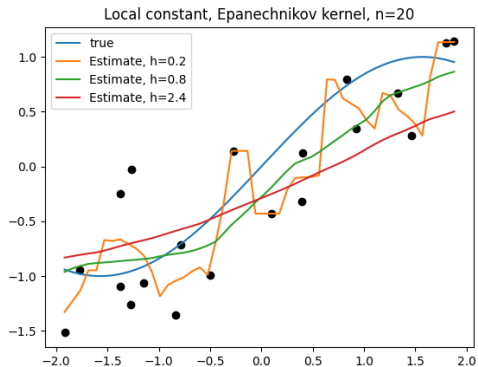
$$\hat{m}(x_0) = \sum_{i=1}^n w_i Y_i$$

where the weights

$$w_i = \frac{k\left(\frac{X_i - x_0}{h}\right)}{\sum_{j=1}^n k\left(\frac{X_j - x_0}{h}\right)}$$

sum up to one. As with the kernel density estimator, we may use kernel functions other than the uniform kernel to avoid abrupt changes in the weights w_i . We call this local constant estimator.

Local Constant (Nadaraya-Watson) Estimator



Local Constant (Nadaraya-Watson) Estimator

We may motivate

$$\hat{m}(x_0) = \frac{\sum_{i=1}^n k\left(\frac{X_i - x_0}{h}\right) Y_i}{\sum_{i=1}^n k\left(\frac{X_i - x_0}{h}\right)}$$

in another way. Notice that we may write $m(x_0)$ as

$$\begin{aligned} m(x_0) = \mathbb{E}[Y_i | X_i = x_0] &= \int_{\mathbb{R}} y f_{Y|X}(y|x_0) dy \\ &= \frac{\int_{\mathbb{R}} y f_{Y,X}(y, x_0) dy}{f_X(x_0)} \end{aligned}$$

where $f_{Y|X}(y|x)$ is the conditional density of Y given X , and $f_{Y,X}(y, x)$ is the joint density of Y and X , and $f_X(x)$ is marginal density of X .

Local Constant (Nadaraya-Watson) Estimator

The kernel density estimator of $f_{Y,X}(y, x)$ and $f_X(x)$ is given by

$$\hat{f}_{Y,X}(y, x) = \frac{1}{nh^2} \sum_{i=1}^n k\left(\frac{X_i - x}{h}\right) k\left(\frac{Y_i - y}{h}\right)$$
$$\hat{f}_X(x) = \frac{1}{nh} \sum_{i=1}^n k\left(\frac{X_i - x}{h}\right)$$

Using these estimators gives the plug-in estimator

$$\begin{aligned}\hat{m}(x) &= \hat{\mathbb{E}}[Y_i | X_i = x] \\ &= \frac{\int_{\mathbb{R}} y \hat{f}_{Y,X}(y, x) dy}{\hat{f}_X(x)}\end{aligned}$$

Local Constant (Nadaraya-Watson) Estimator

Consider the numerator:

$$\begin{aligned}\int_{\mathbb{R}} y \hat{f}_{Y,X}(y, x) dy &= \int_{\mathbb{R}} y \frac{1}{nh^2} \sum_{i=1}^n k\left(\frac{X_i - x}{h}\right) k\left(\frac{Y_i - y}{h}\right) dy \\&= \frac{1}{nh} \sum_{i=1}^n k\left(\frac{X_i - x}{h}\right) \frac{1}{h} \int_{\mathbb{R}} y k\left(\frac{Y_i - y}{h}\right) dy \\&= \frac{1}{nh} \sum_{i=1}^n k\left(\frac{X_i - x}{h}\right) \int_{\mathbb{R}} (Y_i - hu) k(u) du\end{aligned}$$

where we've used the change of variables $u = \frac{Y_i - y}{h}$ with the Jacobian $\left| \frac{dy}{du} \right| = h$. Here,

$$\begin{aligned}\int_{\mathbb{R}} (Y_i - hu) k(u) du &= Y_i \int_{\mathbb{R}} k(u) du - h \int_{\mathbb{R}} uk(u) du \\&= Y_i\end{aligned}$$

Local Constant (Nadaraya-Watson) Estimator

Thus, the plug-in estimator

$$\begin{aligned}\hat{m}(x) &= \hat{\mathbb{E}}[Y_i | X_i = x] \\ &= \frac{\int_{\mathbb{R}} y \hat{f}_{Y,X}(y, x) dy}{\hat{f}_X(x)} \\ &= \frac{\frac{1}{nh} \sum_{i=1}^n k\left(\frac{X_i - x}{h}\right) Y_i}{\frac{1}{nh} \sum_{i=1}^n k\left(\frac{X_i - x}{h}\right)} \\ &= \frac{\sum_{i=1}^n k\left(\frac{X_i - x}{h}\right) Y_i}{\sum_{i=1}^n k\left(\frac{X_i - x}{h}\right)}\end{aligned}$$

is exactly the local constant estimator.

Local Constant (Nadaraya-Watson) Estimator

Suppose we run the regression

$$Y_i = \beta_0 + \varepsilon_i \quad (1)$$

In other words, we find β_0 such that

$$\sum_{i=1}^n (Y_i - \beta_0)^2$$

is minimized. Then, the solution is given by the sample average $\frac{1}{n} \sum_{i=1}^n Y_i$. This regression does not use any information in X_i .

Local Constant (Nadaraya-Watson) Estimator

Suppose in the regression (1), we put the weights $w_i = k \left(\frac{X_i - x_0}{h} \right)$ to each sample Y_i . In other words, we find β_0 such that

$$\sum_{i=1}^n (Y_i - \beta_0)^2 \cdot k \left(\frac{X_i - x_0}{h} \right) \quad (2)$$

is minimized. Then, the solution is given by the weighted sample average

$$\hat{\beta}_0 = \frac{\sum_{i=1}^n k \left(\frac{X_i - x_0}{h} \right) Y_i}{\sum_{i=1}^n k \left(\frac{X_i - x_0}{h} \right)}$$

which is the local constant estimator.

Local Constant (Nadaraya-Watson) Estimator

To see this, note that

$$\sum_{i=1}^n (Y_i - \beta_0)^2 \cdot k\left(\frac{X_i - x_0}{h}\right) = (\mathbf{Y} - \iota_n \beta_0)' \mathbf{K} (\mathbf{Y} - \iota_n \beta_0)$$

where $\mathbf{K} = \text{diag} \left\{ \left(\frac{X_1 - x_0}{h} \right), \dots, \left(\frac{X_n - x_0}{h} \right) \right\}$. The FOC is given by

$$\iota_n' \mathbf{K} (\mathbf{Y} - \iota_n \beta_0) = 0$$

which gives

$$\hat{\beta}_0 = \frac{\iota_n' \mathbf{K} \mathbf{Y}}{\iota_n' \mathbf{K} \iota_n} = \frac{\sum_{i=1}^n k\left(\frac{X_i - x_0}{h}\right) Y_i}{\sum_{i=1}^n k\left(\frac{X_i - x_0}{h}\right)}$$

Local Linear Estimator

Suppose in the regression (2), we use additional regressor $X_i - x_0$. In other words, we find β_0 and β_1 such that

$$\sum_{i=1}^n (Y_i - \beta_0 - \beta_1(X_i - x_0))^2 \cdot k\left(\frac{X_i - x_0}{h}\right)$$

is minimized. Then, we use the value $\hat{\beta}_0$ as the estimate for $\hat{m}(x)$ at $x = x_0$. The reason for including $X_i - x_0$ instead of X_i is so that the intercept equals the conditional expectation at x_0 . We may instead include X_i instead of $X_i - x_0$ and let $\hat{m}(x_0) = \hat{\beta}_0 + \hat{\beta}_1 x_0$.

Local Linear Estimator

The objective function for the local linear estimator is given by

$$\sum_{i=1}^n (Y_i - \beta_0 - \beta_1(X_i - x_0))^2 \cdot k\left(\frac{X_i - x_0}{h}\right) = (\mathbf{Y} - \mathbf{X}\beta)' \mathbf{K} (\mathbf{Y} - \mathbf{X}\beta)$$

where $\mathbf{K} = \text{diag} \left\{ \left(\frac{X_1 - x_0}{h} \right), \dots, \left(\frac{X_n - x_0}{h} \right) \right\}$, and $\mathbf{X}$ is $n \times 2$ matrix collecting ι_n on its first column, and the vector $(X_1 - x_0, \dots, X_n - x_0)'$ on its second column. The FOC is given by

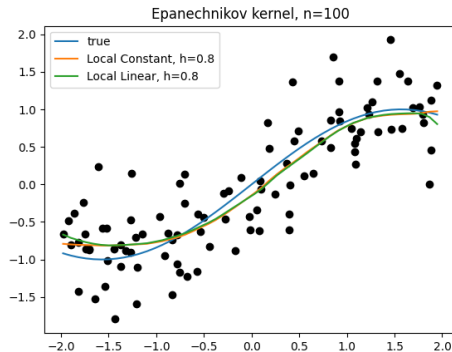
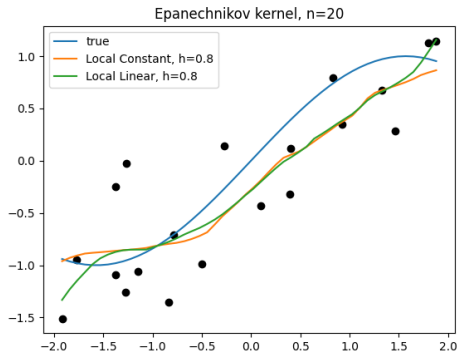
$$\mathbf{X}' \mathbf{K} (\mathbf{Y} - \mathbf{X}\beta) = 0$$

which gives

$$\hat{\beta} = (\mathbf{X}' \mathbf{K} \mathbf{X})^{-1} \mathbf{X}' \mathbf{K} \mathbf{Y}$$

and our estimator $\hat{m}(x_0)$ is the first element of $\hat{\beta}$.

Local Linear Estimator



While local linear estimator has better statistical properties (especially at the boundary), it does not always beat local constant estimator.

What if we have $X_i \in \mathbb{R}^k$?

All the previous arguments still applies with the univariate kernel function $k : \mathbb{R} \rightarrow \mathbb{R}$ replaced with a multivariate kernel function $K : \mathbb{R}^k \rightarrow \mathbb{R}$ and the local constant estimator given as

$$\hat{m}(x) = \frac{\sum_{i=1}^n K\left(\frac{X_i - x}{h}\right) Y_i}{\sum_{i=1}^n K\left(\frac{X_i - x}{h}\right)}$$

and the local linear estimator given as the minimizer $\hat{\beta}_0$ of

$$\sum_{i=1}^n (Y_i - \beta_0 - \beta_1(X_i - x_0))^2 \cdot K\left(\frac{X_i - x_0}{h}\right)$$

However, the estimation result deteriorates very quickly as the dimensionality k increases. This is called the *curse of dimensionality*.